Minh Le

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Second-year mechanical engineering student passionate about sustainable technology and hands-on fabrication. Eager to turn theoretical learning into real, practical builds for electric and solar vehicle systems.

Education

University of British Columbia, Vancouver — **Expected graduation: May 2029**

BASc, Mechanical Engineering · Second Year · Cumulative GPA: 4.00/4.33 (A) · Dean's List

Projects

Two-Speed Hand-Crank Generator — **June 2026 – August 2026**

Personal project

- Designed and built a **custom-made** hand-crank generator that sets its output voltage through gear selection instead of regulator electronics, **charging a phone** from hand power alone.
- Built a two-speed gearbox with a constant-mesh dog clutch that shifts under load; proved with logged data that each gear selects a distinct voltage (high gear ~1.9× low), modeled in **SolidWorks** and **3D printed** in PLA with **hand-machined** shafts.
- Characterized the **hand-wound axial-flux generator** with an **Arduino**, **INA219**, and **Hall effect sensor**; measured drivetrain efficiency up to ~**48%** around 42-52 crank RPM.

Leader-Follower Robotic Arm — **May 2026 – June 2026**

Personal project

- Designed and built a 4-DOF arm that mirrors a hand-held twin in real time, with record and playback. Both arms were modelled in **SolidWorks** and **3D printed** in PLA.
- Designed a custom **Arduino shield PCB** in **KiCad**, manufactured by **JLCPCB**.
- Wrote three-mode firmware with sensor smoothing; fixed voltage dips and swapped to metal-gear servos for load.

Rainwater Harvester: System Simulation & Design in Excel — **March 2026 – April 2026**

School project · 5-person team

- Built the topography model and a **5-year** daily water-balance **simulation** tracking system reliability.
- Modeled on-demand flow and pump-to-storage computationally from pipe friction, elevation, and pump curves.
- Placed **2nd of 10** in studio at **63.66% satisfaction** on the coordinators' withheld-weather simulation.

Technical Skills

Design/CAD: SolidWorks, Fusion 360, PrusaSlicer (slicing software), technical drawing (orthographic, isometric)
Electronics: PCB design in KiCad (schematic, board layout, trace sizing), Arduino, soldering, basic circuits & sensors
Programming: Embedded C/C++ (Arduino), Python, MATLAB, Git/GitHub, VS Code
Fabrication: 3D printing (FDM), manual fabrication, sheet-metal fabrication
Productivity/Data: Excel-based system modeling & data analysis, Microsoft Office, Google Workspace

Job Experience

Volleydome — **May–Aug 2026, Jan 2024 – Jun 2025**

Club Volleyball Coach — *Calgary, AB*

- **Coached** groups of 20-30 athletes, grades 6-11, adapting practice plans from fundamentals to skill refinement.
- Gave **individual feedback** and **communicated** with athletes, parents, and staff, across multiple seasons.

Volleydome — **Mar 2023 – Dec 2023**

Volunteer — *Calgary, AB*

- Supported event setup, managed logistics, and helped participants with questions throughout the day.

Additional

Languages: English · Availability: Full Availability · Driver's licence: Class 5 (Alberta)